

EJERCICIOS 2 COMPLEJOS

DIEGO CONTRERAS

①

$$\begin{cases} \frac{dn_h}{dt} = \alpha n_h - \beta n_h n_e - \epsilon n_h^2 \\ \frac{dn_e}{dt} = \delta n_h n_e - \gamma n_e \end{cases} \quad \alpha, \beta, \epsilon, \delta, \gamma > 0$$

a) Fixed point $\rightarrow \frac{dn_h}{dt} = \frac{dn_e}{dt} = 0 \Rightarrow$

$$\begin{cases} \alpha n_h - \beta n_h n_e - \epsilon n_h^2 = 0 \\ \delta n_h n_e - \gamma n_e = 0 \end{cases} \rightarrow \begin{cases} n_h (\alpha - \beta n_e - \epsilon n_h) = 0 \\ n_e (\delta n_h - \gamma) = 0 \end{cases}$$

Trivial solution $\rightarrow n_h = n_e = 0$. We don't want it. $\Rightarrow$

$$\begin{cases} \alpha - \beta n_e - \epsilon n_h = 0 \\ \delta n_h - \gamma = 0 \end{cases} \rightarrow \begin{cases} \boxed{n_h = \frac{\gamma}{\delta}} \\ \boxed{n_e = \frac{\alpha - \beta \frac{\gamma}{\delta}}{\beta}} \end{cases} \rightarrow \text{Non trivial solutions.}$$

But $n_h, n_e > 0 \Rightarrow \alpha - \frac{\epsilon \gamma}{\delta} > 0 \rightarrow \alpha > \frac{\epsilon \gamma}{\delta} \Rightarrow \boxed{\alpha > \epsilon n_h}$

b)

$$J_1 = \alpha n_h n_e - \rho n_h n_e - \epsilon n_h^2;$$

$$J_2 = \delta n_h n_e - \delta n_e$$

$$J = \begin{pmatrix} \frac{\partial J_1}{\partial n_h} & \frac{\partial J_1}{\partial n_e} \\ \frac{\partial J_2}{\partial n_h} & \frac{\partial J_2}{\partial n_e} \end{pmatrix}$$

$$\frac{\partial J_1}{\partial n_h} = \alpha - \rho n_e - 2\epsilon n_h;$$

$$\frac{\partial J_1}{\partial n_e} = -\rho n_h;$$

$$\frac{\partial J_2}{\partial n_h} = \delta n_e;$$

$$\frac{\partial J_2}{\partial n_e} = \delta n_h - \delta$$

$$J = \begin{pmatrix} \alpha - \rho n_e - 2\epsilon n_h & -\rho n_h \\ \delta n_e & \delta n_h - \delta \end{pmatrix} \quad \begin{cases} n_h^* = \frac{\delta}{\epsilon} \\ n_e^* = \frac{\alpha - \epsilon \delta / \epsilon}{\rho} \end{cases}$$

$$\Rightarrow \alpha - \rho n_e^* - 2\epsilon n_h^* = \alpha - (\alpha - \epsilon \delta / \epsilon) - 2\epsilon \frac{\delta}{\epsilon} = -\epsilon \frac{\delta}{\epsilon}$$

$$-\rho n_h^* = -\rho \frac{\delta}{\epsilon}$$

$$\delta n_e^* = \frac{\alpha \delta - \epsilon \delta}{\rho}$$

$$\delta n_h^* - \delta = 0$$

$\Rightarrow$

$$J^* = \begin{pmatrix} -\epsilon \frac{\delta}{\epsilon} & -\rho \frac{\delta}{\epsilon} \\ \frac{\alpha \delta - \epsilon \delta}{\rho} & 0 \end{pmatrix}$$

$n_h^* \rightarrow n_e^*$

c)

$$Z = \text{Trace}(J^*) = \left[\frac{-\epsilon\delta}{\delta} \right]$$

$$\Delta: \det(J^*) = \frac{\beta\delta}{\delta} \left(\frac{\alpha\delta - \epsilon\delta}{\delta} \right) = \alpha\delta - \frac{\epsilon\delta^2}{\delta}$$

Do we need the determinant to be positive?

Let's diagonalize J^* ;

$$\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc = 0; \quad \lambda^2 - \lambda \overbrace{(a+d)}^Z - \underbrace{bc}_{+\Delta} = 0$$

$$\Rightarrow \lambda^2 - Z\lambda + \Delta = 0$$

$$\Rightarrow \text{We need } \text{Re}(\lambda) < 0 \rightarrow |Z| > \sqrt{Z^2 - 4\Delta}$$

$$\Rightarrow \lambda = \frac{Z \pm \sqrt{Z^2 - 4\Delta}}{2}$$

$$\rightarrow \text{For } \lambda \text{ to be real } Z^2 - 4\Delta \geq 0$$

We know for a fact (all ctes are positive) that the trace is negative: $Z < 0$,

$\rightarrow$ For it to be a stable node we need λ to be real, so $Z^2 - 4\Delta > 0$

because Z is negative $\Rightarrow$

Δ is positive
or
negative

$$\text{and } Z > \sqrt{Z^2 - 4\Delta}$$

if Δ is positive, $|Z| > 4\Delta$

$\rightarrow$ For it to be a stable spiral or ~~stable~~ (sink spiral) we need all real parts negative

$$\text{and have imaginary part } \rightarrow Z^2 - 4\Delta < 0$$

d) Now to calculate $\tau^2 - 4\epsilon\Delta$ ϵ

$$\tau^2 = \left(-\epsilon \frac{\partial}{\partial} \right)^2 = \epsilon^2 \frac{\partial^2}{\partial^2} ;$$

$$\Delta = \alpha\delta - \epsilon \frac{\partial^2}{\partial}$$

$$\tau^2 - 4\Delta = \frac{\epsilon^2 \delta^2}{\sigma^2} - 4\alpha\delta + 4\epsilon \frac{\partial^2}{\partial} ;$$

$$\rightarrow \text{Ecosystem A: } \tau^2 - 4\Delta \Big|_A = \frac{1^2 \cdot 1^2}{1^2} - 4 \cdot 2 \cdot 1 + 4 \cdot 1 \cdot \frac{1}{1} = 1 - 8 + 4 = -3 < 0$$

$\Rightarrow$ We have a spiral.

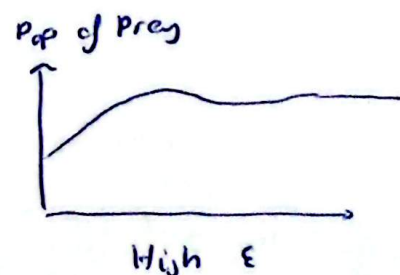
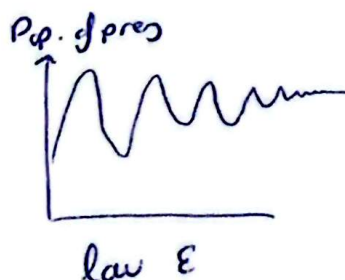
$$\rightarrow \text{Ecosystem B: } \tau^2 - 4\Delta \Big|_B = \frac{3^2 \cdot 1}{1} - 4 \cdot 2 \cdot 1 + 4 \cdot 3 \cdot \frac{1}{1} = 9 - 8 + 12 = 13 > 0$$

$\Rightarrow$ We have a stable node.

e) Biological implication?

ϵ defines the ratio for "competition for grass", so the deaths of the prey not caused by the predator (so it doesn't increase their population).

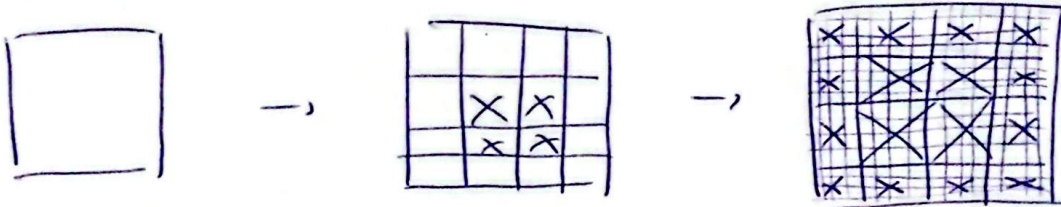
If we increase ϵ what we obtain is the equilibrium point is achieved more "monotonous". I mean, it does not fluctuate that a lot, it gets "straight to the point".



(but remember, $i \propto \frac{\partial}{\partial}$!!)

②

We take a square and divide it in 16. We remove the 4 central parts:



We calculate the fractal dimension with $D = \lim_{h \rightarrow \infty} \frac{\log(M_h)}{\log(\frac{1}{L_h})}$;

So when we divide and remove we go to:

$$10^0 \rightarrow 10^1 \rightarrow 10^2$$

$$N = 1$$

$$N^* = 16 - 4 = 12$$

$$N = 12 \cdot (16 - 4) = 12^2 = 144$$

$$A = 1$$

$$A = \frac{1}{4}$$

$$A = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} = \frac{1}{4^2}$$

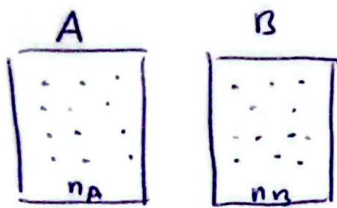
So in the h iteration we have 12^h squares and each square occupies $\frac{1}{4^h}$ of the area.

$$N_h = 12^h; \quad L_h = \frac{1}{4^h}; \quad \rightarrow \frac{1}{L_h} = 4^h,$$

$$D = \lim_{h \rightarrow \infty} \frac{\log(12^h)}{\log(\frac{1}{4^h})} = \lim_{h \rightarrow \infty} \frac{h \log(12)}{h \log(4)} = \frac{\log(12)}{\log(4)} = 1.79248 = D$$

For the numerical solution we have a notebook in jupyter.

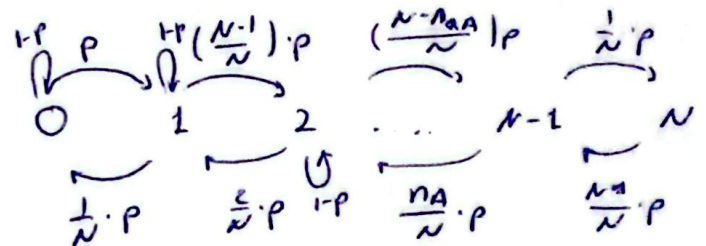
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→ Choose one at random.

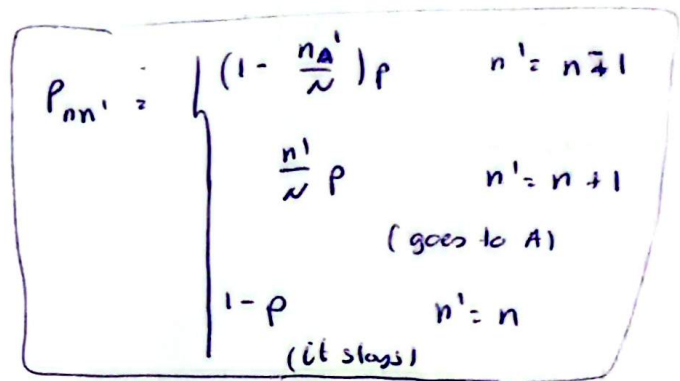
→ Probability p to move to the other box, if not, it stays.

I will call $N = n_A + n_B$; $n_A =$



Notation: $P_{nn'} = P(n' | n)$

$n' \equiv n_A$



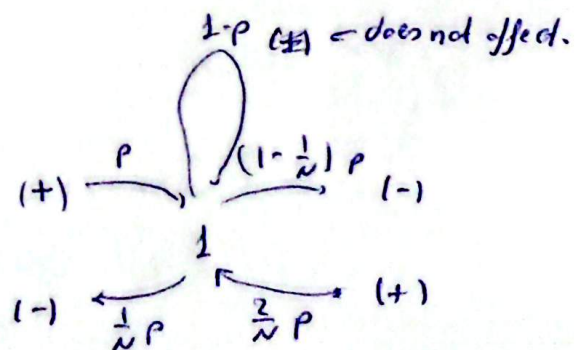
Markov Chain:

→ Master equation:

$$\frac{dP(x,t)}{dt} = \int [w(x|y)P(y,t) - w(y|x)P(x,t)] dy \quad (\text{continuum})$$

$$= \sum_{n'} (P_{nn'} P_{n'} - P_{n'n} P_n)$$

for example for the case of 1:



" p " probability of moving

" p " does not affect the dynamics, only the time scale at which the system evolves.

~~Processes that enter:~~

$$\frac{dP(n,t)}{dt} = \text{enter} - \text{exit}:$$

Processes that enter: from $n-1 \rightarrow n$, from $n+1 \rightarrow n$

Processes that exit: from $n \rightarrow n-1$; from $n \rightarrow n+1$;

$$\begin{aligned} \frac{dP(n,t)}{dt} = & w(n|n-1)P(n-1,t) + w(n|n+1)P(n+1,t) \\ & - w(n+1|n)P(n,t) - w(n-1|n)P(n,t) \end{aligned}$$

Now term by term:

$$\begin{aligned} w(n|n-1) &= p \cdot \frac{N-n+1}{N} ; & w(n|n+1) &= p \cdot \frac{n+1}{N} \\ w(n+1|n) &= p \cdot \frac{n+1}{N} ; & w(n-1|n) &= p \cdot \frac{n}{N} \end{aligned} \quad \left\{ \begin{array}{l} \text{weights} \end{array} \right.$$

~~$P(n,t)$~~ ~~$P(n,t)$~~

$$\frac{dP(n,t)}{dt} = p \cdot \frac{N-n+1}{N} P(n-1,t) + p \cdot \frac{n+1}{N} P(n+1,t) - p \cdot \frac{n+1}{N} P(n,t) - p \cdot \frac{n}{N} P(n,t)$$

$$= p \cdot \frac{N-n+1}{N} P(n-1,t) + p \cdot \frac{n+1}{N} P(n+1,t) - p P(n,t) = \frac{dP(n,t)}{dt}$$

MASTER EQUATION

(for $1 \leq n \leq N-1$)

But for $n=0$ and $n=N$ there is not going back nor forward respectively.

For $n=0$: There is no $n-1$.

$$\left. \frac{dP(0,t)}{dt} = p \frac{n+1}{N} P(n+1,t) - p \frac{N-n}{N} P(n,t) \right|_{n=0} = \boxed{p \frac{1}{N} P(1,t) - p P(0,t) = \frac{dP(0,t)}{dt}}$$

For $n=N$: There is no $n+1$;

$$\left. \frac{dP(N,t)}{dt} = p \frac{N-n+1}{N} P(n-1,t) - p \frac{n}{N} P(n,t) \right|_{n=N} = \boxed{p \frac{1}{N} P(N-1,t) - p P(N,t) = \frac{dP(N,t)}{dt}}$$

•) Steady-state probabilities:

For a steady-state we need $\frac{dP(n,t)}{dt} = 0 \quad \forall n,$

For $n \in [1, N-1]$:

~~$$\frac{dP}{dt} = 0 = p \frac{N-n+1}{N} P(n-1,t) + p \frac{N+1}{N} P(n+1,t) - p P(n,t)$$~~

It is better to do a detail balance and do it via flux. The system will be steady if the amount of increasing is the same as decreasing in each node.

From $n \rightarrow n+1$: $P(n) W(n|n+1)$

From $n+1 \rightarrow n$: $P(n+1) W(n+1|n)$

} we equal these.

$$P(n) W(n+1|n) = P(n+1) W(n|n+1),$$

$$P(n) \left(p \frac{N-n}{N} \right) = P(n+1) \left(p \frac{n+1}{N} \right);$$

$$P(n) (N-n) = P(n+1) (n+1); \quad \rightarrow \quad \boxed{P(n+1) = P(n) \cdot \frac{N-n}{n+1}}$$

So we have this recurrent equation

start with 0 and continue to see the pattern;

$$P(1) = P(0) \frac{N}{1}; \quad P(2) = P(1) \cdot \frac{N-1}{2} = P(0) \frac{N}{1} \cdot \frac{(N-1)}{2},$$

$$P(3) = P(2) \frac{N-2}{3} = P(0) \frac{N}{1} \frac{(N-1)}{2} \frac{(N-2)}{3} = P(0) \frac{N(N-1)(N-2)}{3!}$$

~~For the case of n~~ For the n case:

$$\begin{aligned} P(n) &= P(0) \cdot \frac{N(N-1)(N-2)\dots(N-n+1)}{n!} = P(0) \frac{N!}{n! (N-n)!} \\ &= P(0) \frac{N!}{n!} \cdot \frac{(N-n)(N-n-1)\dots(N-n+1)}{(N-n)!} \\ &= P(0) \cdot \frac{N!}{n! (N-n)!} = \boxed{P(0) \binom{N}{n} = P(n)} \end{aligned}$$

This is not over, we have to normalize. We will calculate $P(0)$ so that it

fulfills this normalization.

·) Normalization:

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$$\sum_{n=0}^{\infty} P(n) = 1 \Rightarrow \sum_{n=0}^{\infty} P(0) \binom{N}{n} = P(0) \sum_{n=0}^{\infty} \binom{N}{n},$$

$$\left((a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k; \quad a=b=1 \Rightarrow (2)^N = \sum_{n=0}^{\infty} \binom{N}{n} 1^{N-n} \cdot 1^n \right)$$

$$\Rightarrow \sum_{n=0}^{\infty} P(n) = 2^N \cdot P(0) = 1 \rightarrow P(0) = \frac{1}{2^N} = 2^{-N}$$